AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 14 · WORKBOOK

Hallucination Failure Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected claim-verification record.

Explain the procedure, trace statuses, audit missing evidence and design a revealing check.

Keep one habit: Retain original claims, passages, calculations, status words and source-based repairs.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Multimodal Evidence

Visual Composition

Structured Output and Schemas

W1 · Explain claim-level verification

> Name each claim, source passage and calculation if needed.

Your evidence record

- Main room review: R says B seats12,meeting15:00;answer says20,15:00,free lunch. Capacity contradicted,time supported,lunch unsupported. Separate trip S:12km,constant4km/h,no stops,departure09:00;draft2hours,09:00,lunch. Calculate moving hours from S,not R;30-minute stop is an added planned condition. No lunch source or real observed trip supplied.

- Reference for this exercise: open this week's student lesson, "Hallucination Failure Types", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the room-report audit

> Keep supported, contradicted and unsupported claims distinct.

Your evidence record

- Main room review: R says B seats12,meeting15:00;answer says20,15:00,free lunch. Capacity contradicted,time supported,lunch unsupported. Separate trip S:12km,constant4km/h,no stops,departure09:00;draft2hours,09:00,lunch. Calculate moving hours from S,not R;30-minute stop is an added planned condition. No lunch source or real observed trip supplied.

- Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit missing evidence and arithmetic

> A missing fact does not establish its opposite.

Your evidence record

- Main room review: R says B seats12,meeting15:00;answer says20,15:00,free lunch. Capacity contradicted,time supported,lunch unsupported. Separate trip S:12km,constant4km/h,no stops,departure09:00;draft2hours,09:00,lunch. Calculate moving hours from S,not R;30-minute stop is an added planned condition. No lunch source or real observed trip supplied.

- A report omits this qualification: Source agreement and arithmetic validity are separate checks; missing **evidence** is not proof of the opposite claim.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing claim check

> Preserve sources, assumptions and the expected status before testing.

Your evidence record

- Main room review: R says B seats12,meeting15:00;answer says20,15:00,free lunch. Capacity contradicted,time supported,lunch unsupported. Separate trip S:12km,constant4km/h,no stops,departure09:00;draft2hours,09:00,lunch. Calculate moving hours from S,not R;30-minute stop is an added planned condition. No lunch source or real observed trip supplied.

- Reference for this exercise: open this week's student lesson, "Hallucination Failure Types", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit a new trip summary independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use the independent source's own distance, constant speed and exclusion of stops. Show the calculation's units and state which claims have direct, calculated or missing support. No actual trip observations or lunch record are supplied; do not import values from the worked trip or invent real conditions.

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Generation failures and verification. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 14 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

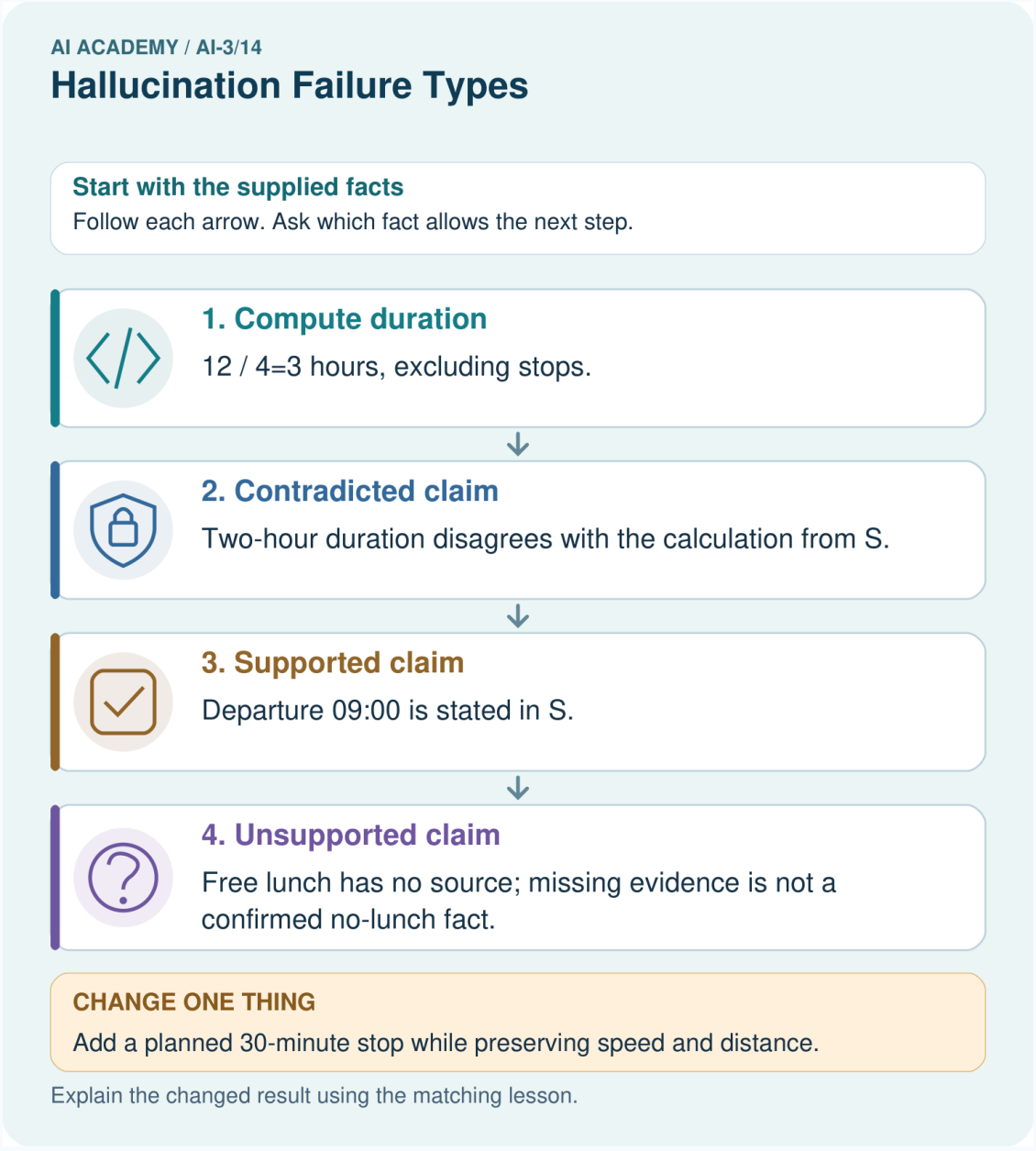
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy drafts a field-trip summary from one approved record.

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.



Follow the labelled picture with the matching explanation. Open the vector diagram

Can we conclude there will definitely be no lunch from S alone?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy drafts a field-trip summary from one approved record.
- Which labelled step matches this detail? Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.
- 1. **Unsupported claim**
- 2. Supported claim

- 3. **Contradicted** claim
- 4. Compute duration
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Unsupported claim** Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Different errors need different repairs.

Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Changed-case question

- Changed-case practice: Add a planned 30-minute stop while preserving speed and distance. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Planned total elapsed time becomes 3.5 hours.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Repair and save your work

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” **unsupported** before opening the box?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

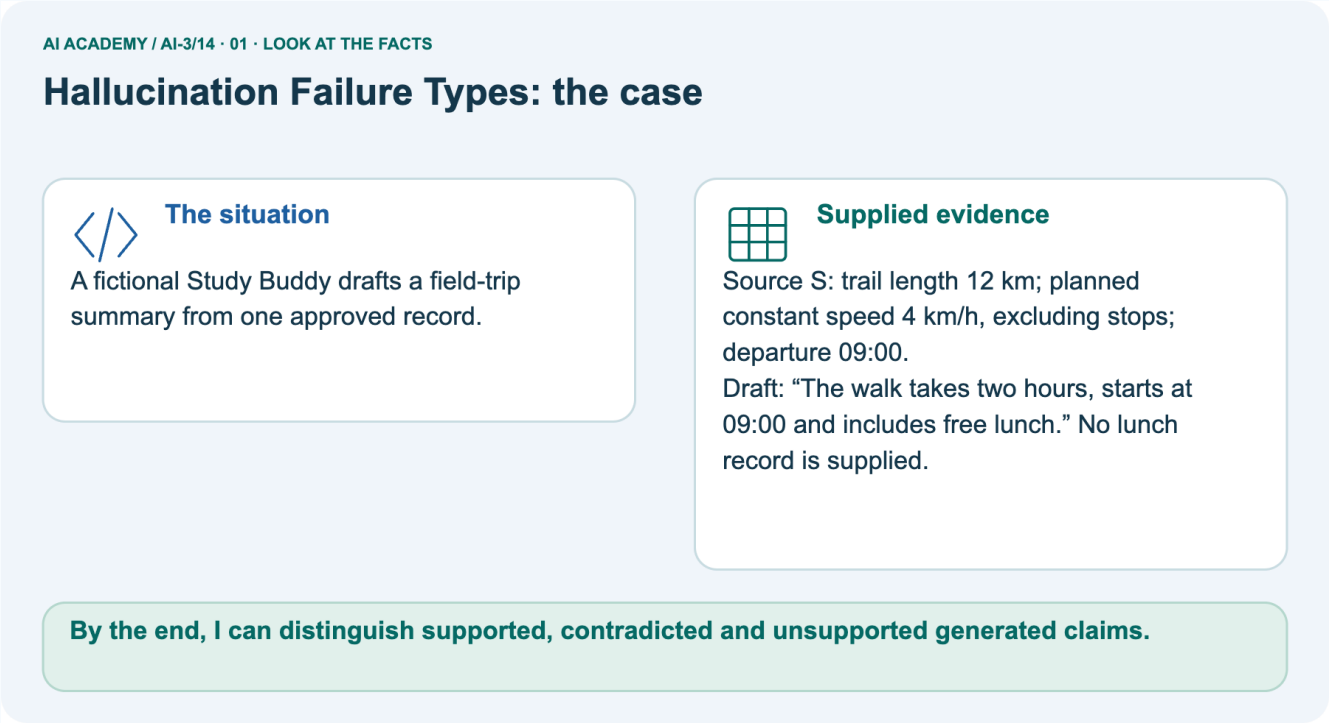
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Hallucination Failure Types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

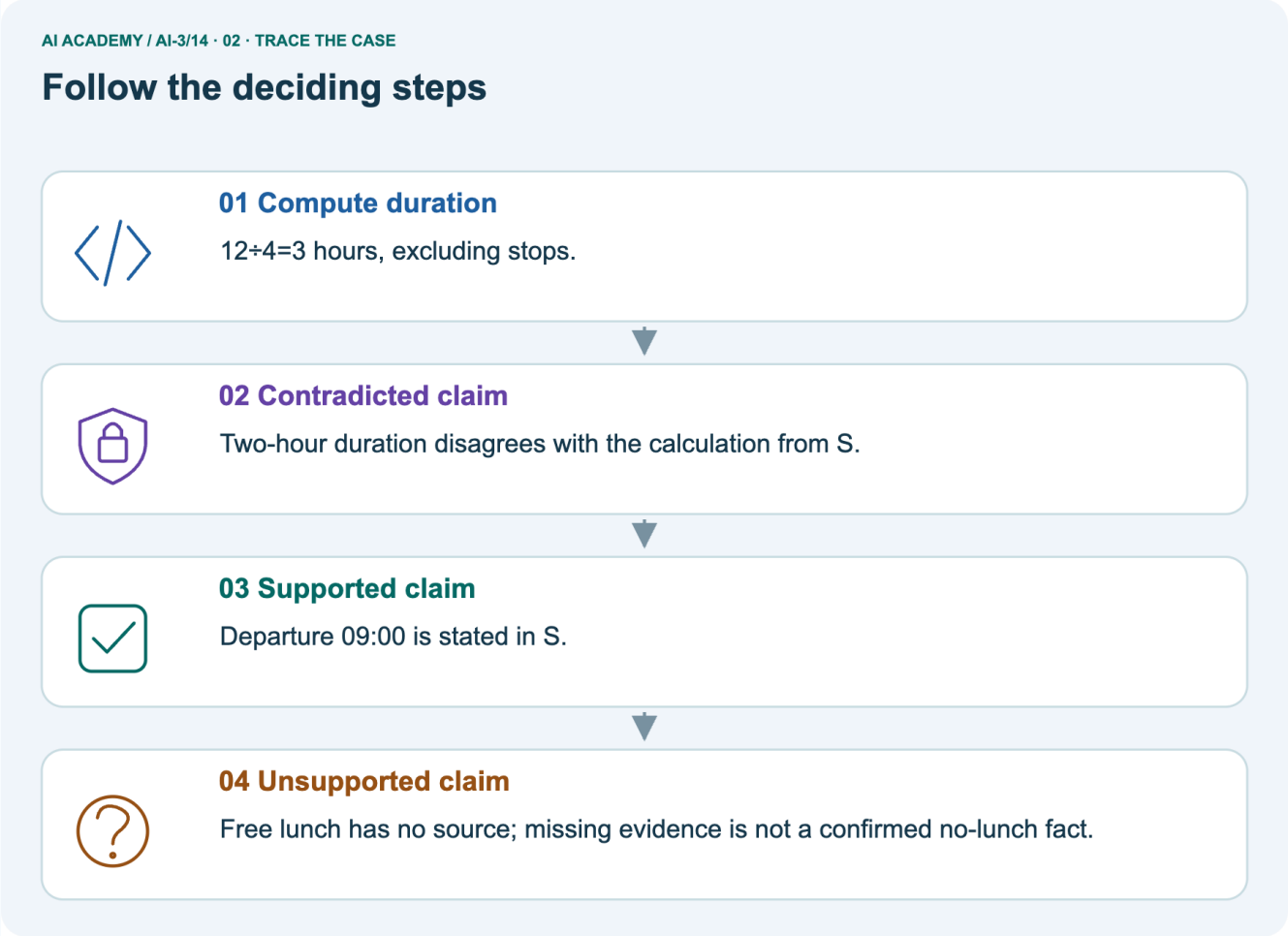
A fictional Study Buddy drafts a field-trip summary from one approved record. Supplied facts: Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



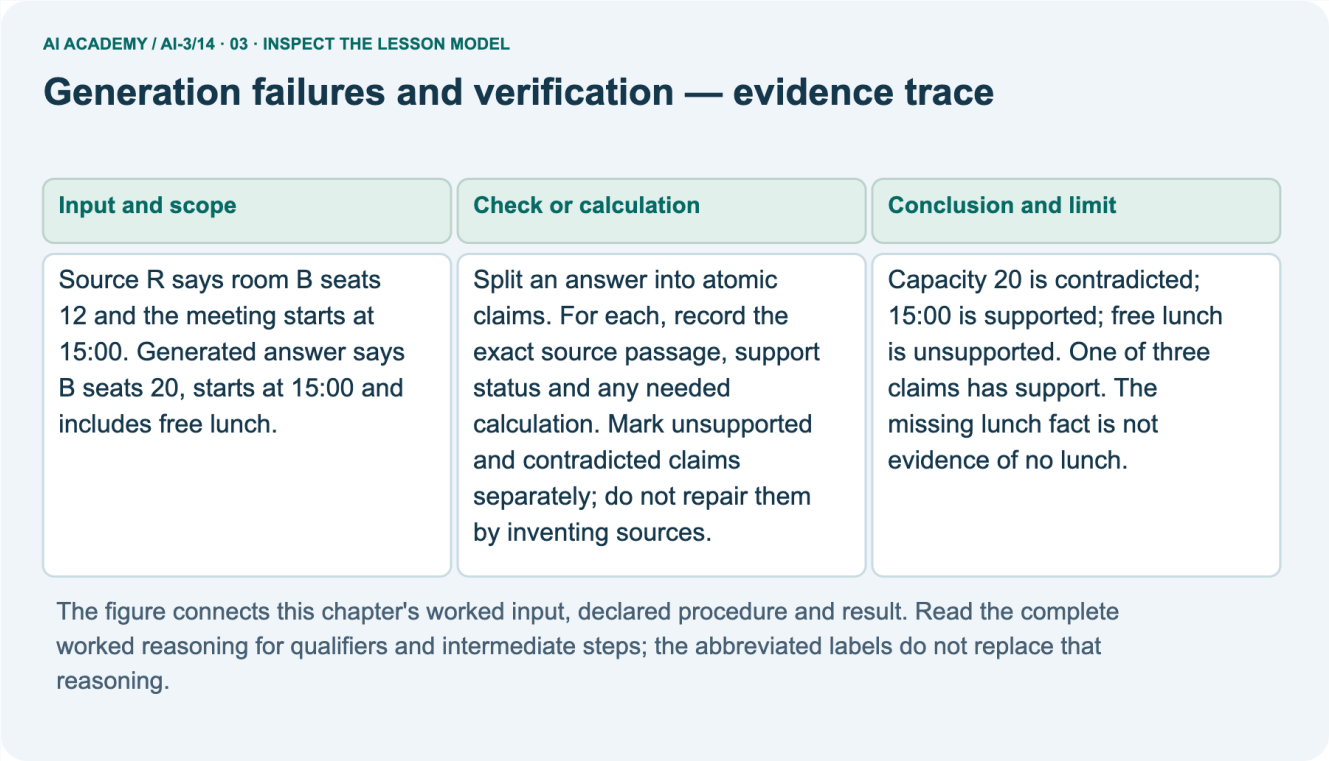
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute duration: 12÷4=3 hours, excluding stops. → **Contradicted** claim: Two-hour duration disagrees with the calculation from S. → Supported claim: Departure 09:00 is stated in S. → **Unsupported claim**: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Generation failures and verification — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generation failures and verification — **evidence** trace. Source R says room B seats 12 and the meeting starts at 15:00.

Generated answer says B seats 20, starts at 15:00 and includes free lunch. | Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation.

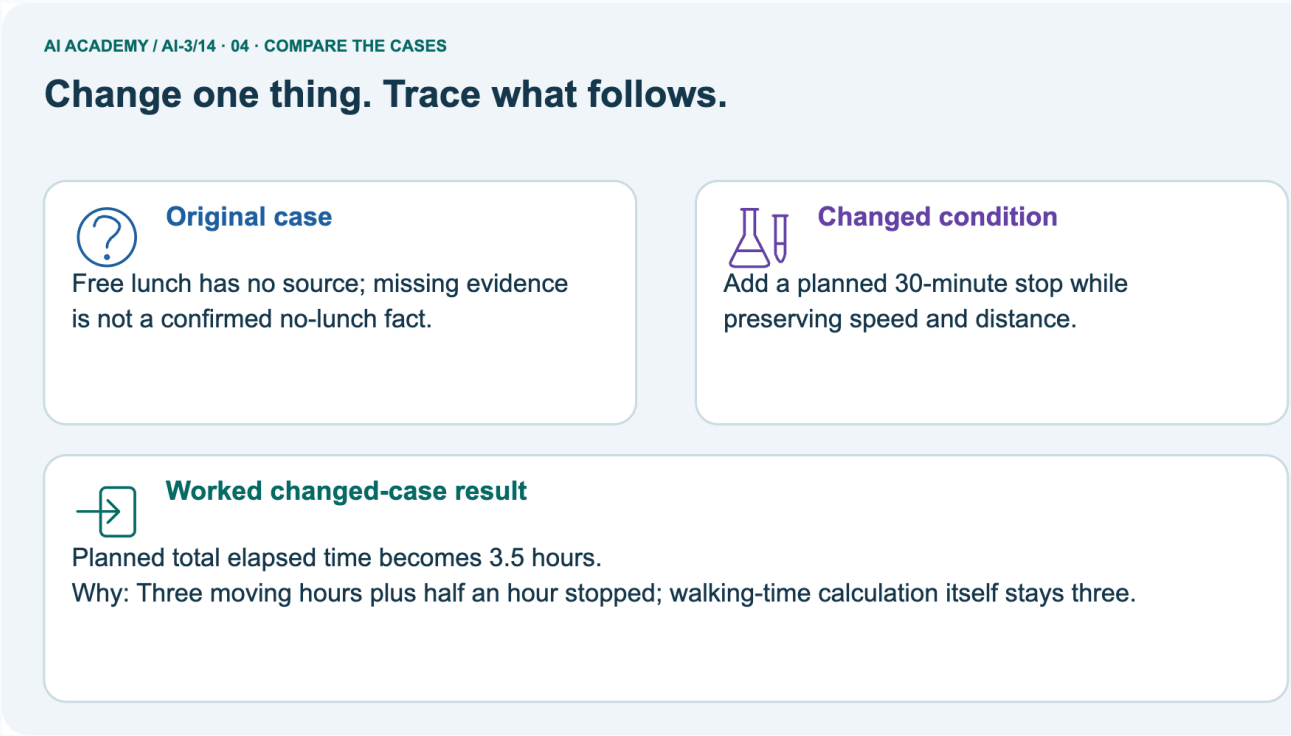
Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources. | Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support.

The missing lunch fact is not **evidence** of no lunch.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

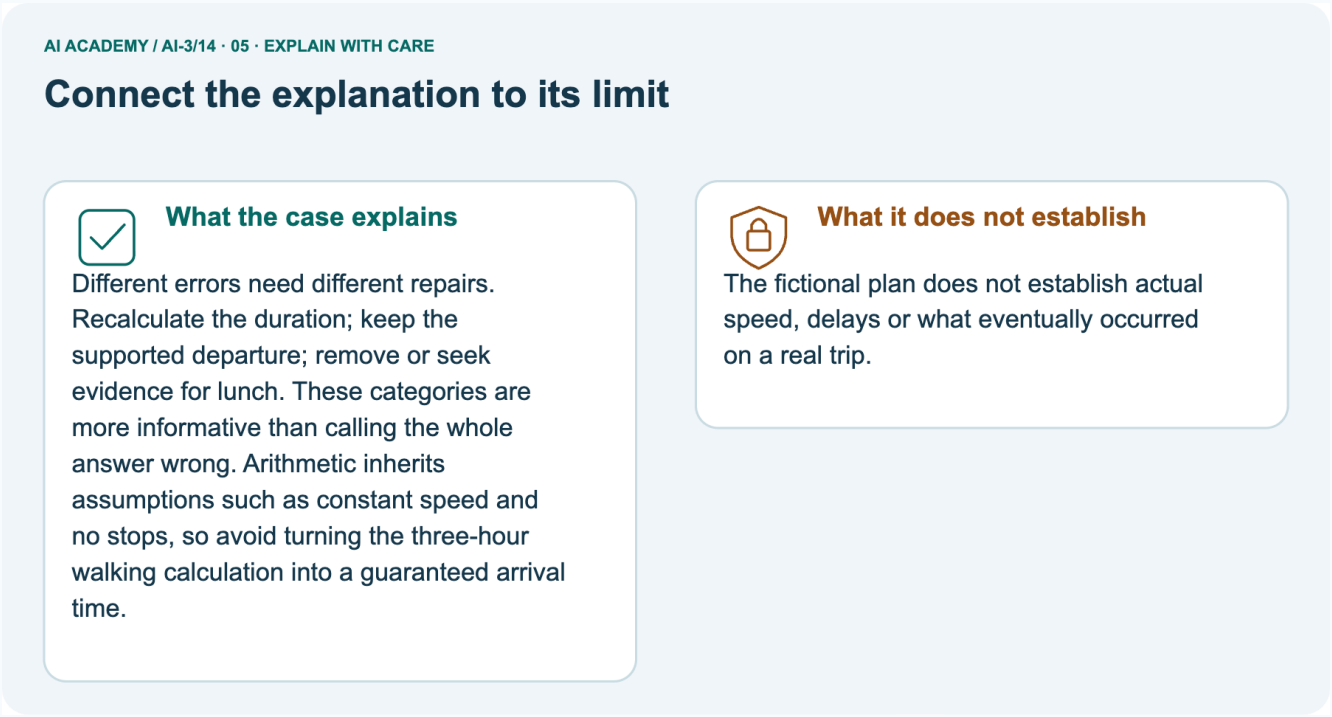
Original case: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.
Changed condition: Add a planned 30-minute stop while preserving speed and distance.

Expected result: Planned total elapsed time becomes 3.5 hours. Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

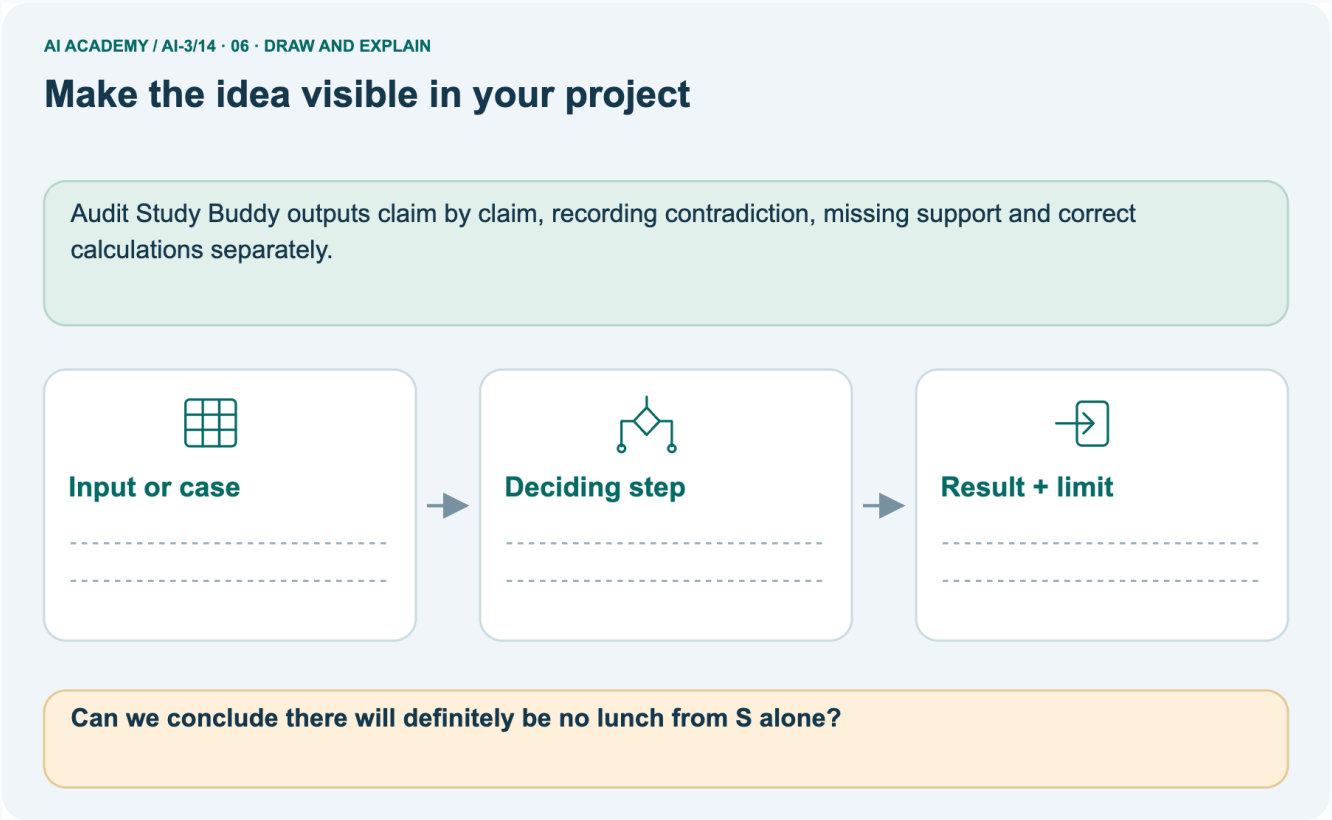
Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Limit: The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately. Draw your **input**, deciding step, result and limit.

Supported question: Can we conclude there will definitely be no lunch from S alone?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Hallucination Failure Types: the case

02. Follow the deciding steps

03. Generation failures and verification — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.